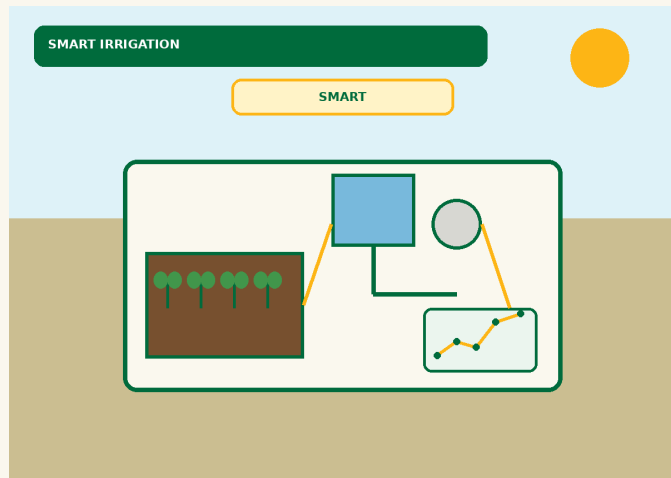




SMART

SMART IRRIGATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

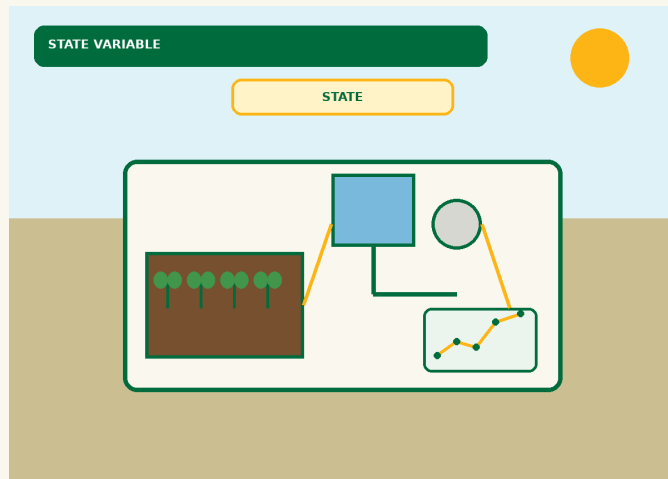
TECHNICAL CONCEPT

Smart irrigation uses measurements, rules, models, or predictions to adapt water application to crop and system conditions.

ANALYSIS / DESIGN TASK

Define the information-to-action pathway.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



STATE

STATE VARIABLE

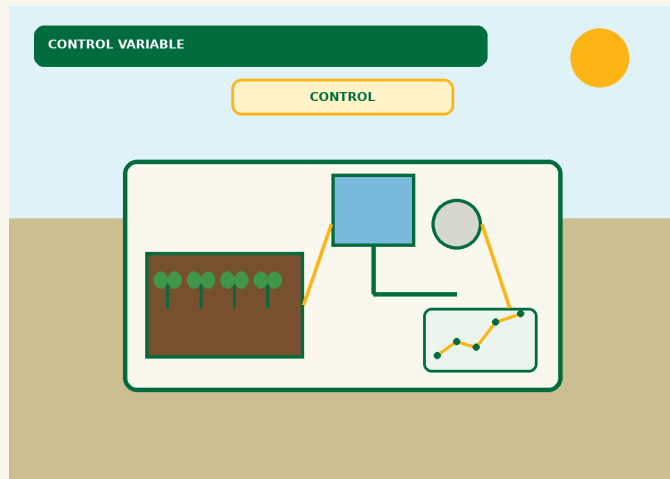
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A state variable describes a system condition needed to represent current status, such as soil water, tank level, or crop stage.

ANALYSIS / DESIGN TASK

Identify three state variables.



CONTROL

CONTROL VARIABLE

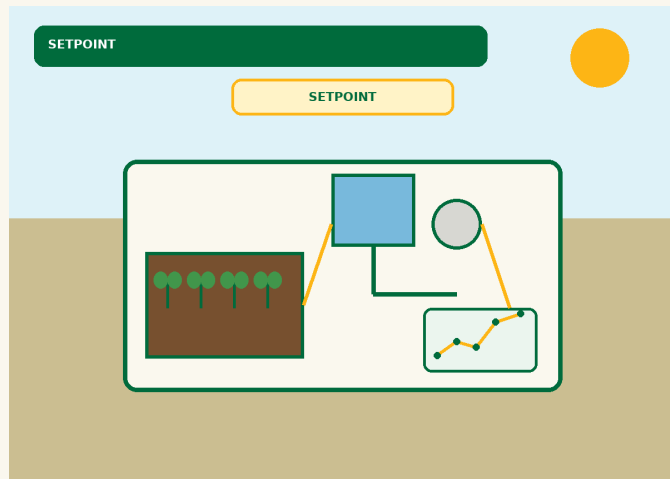
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A control variable is an adjustable operating decision such as valve state, irrigation duration, or flow setpoint.

ANALYSIS / DESIGN TASK

Define one controllable action.



SETPOINT

SETPOINT

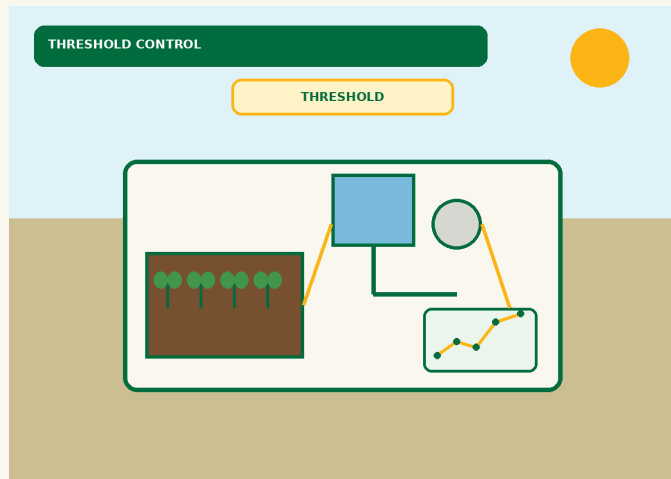
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A setpoint is the target value that a control system attempts to maintain or achieve.

ANALYSIS / DESIGN TASK

Choose a soil-moisture or tank-level setpoint.



THRESHOLD

THRESHOLD CONTROL

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

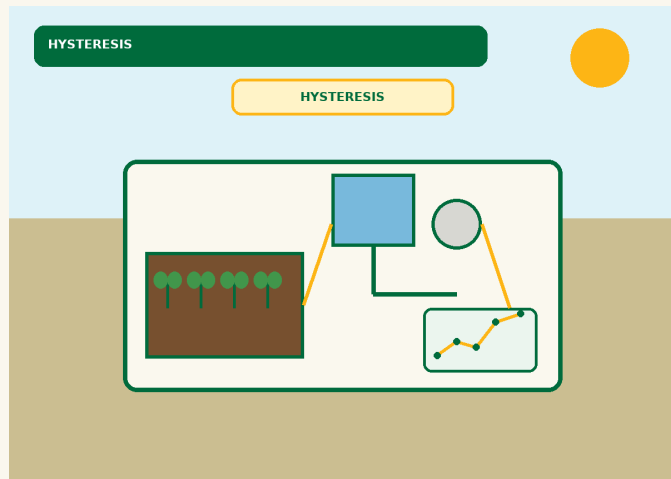
TECHNICAL CONCEPT

Threshold control triggers an action when a measured variable crosses a defined limit.

ANALYSIS / DESIGN TASK

Write an irrigation on/off rule.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



HYSTERESIS

HYSTERESIS

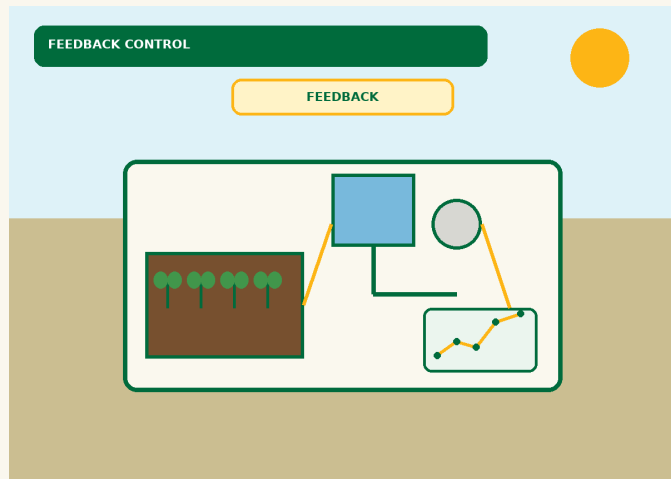
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Hysteresis uses different on and off thresholds to reduce rapid switching around a single boundary.

ANALYSIS / DESIGN TASK

Design two moisture thresholds.



FEEDBACK

FEEDBACK CONTROL

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

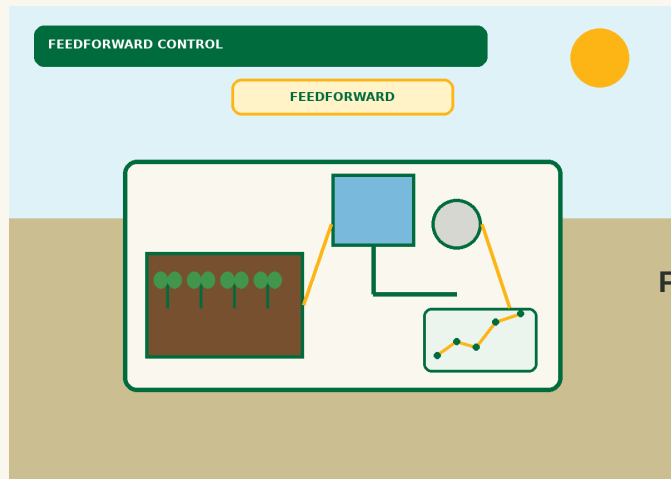
TECHNICAL CONCEPT

Feedback control measures system response and uses that information to adjust later control action.

ANALYSIS / DESIGN TASK

Draw a sensor-controller-valve feedback loop.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



FEEDFORWARD

FEEDFORWARD CONTROL

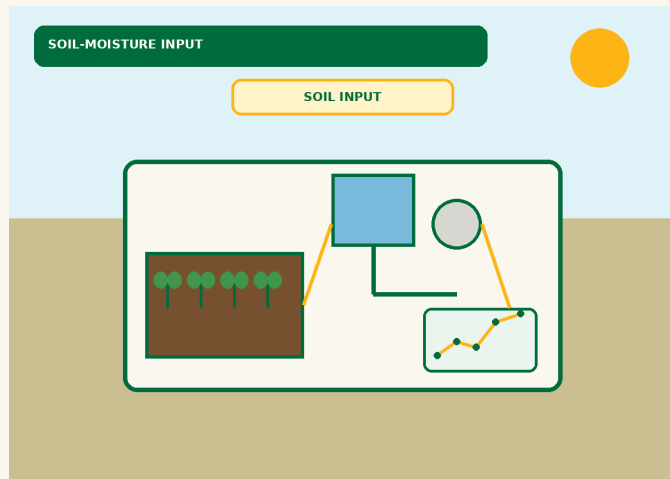
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Feedforward control anticipates a disturbance or demand using information measured before the controlled variable changes.

ANALYSIS / DESIGN TASK

Give an example using expected water demand.



SOIL INPUT

SOIL-MOISTURE INPUT

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

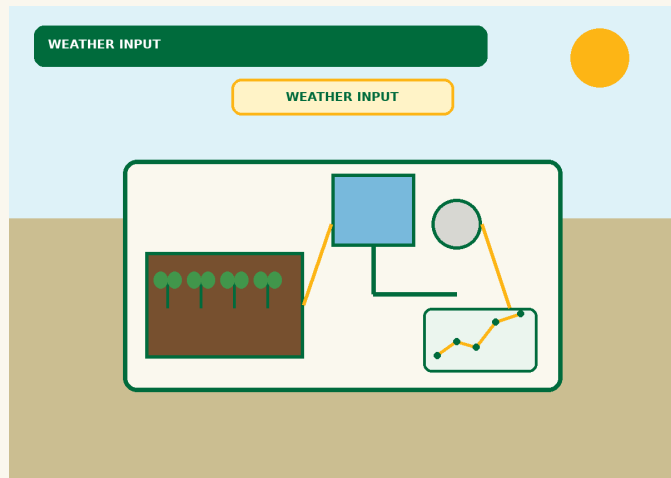
TECHNICAL CONCEPT

Soil-moisture measurements can inform irrigation decisions when sensor placement, calibration, rooting depth, and thresholds are appropriate.

ANALYSIS / DESIGN TASK

Evaluate two sensor readings before irrigation.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



WEATHER INPUT

WEATHER INPUT

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

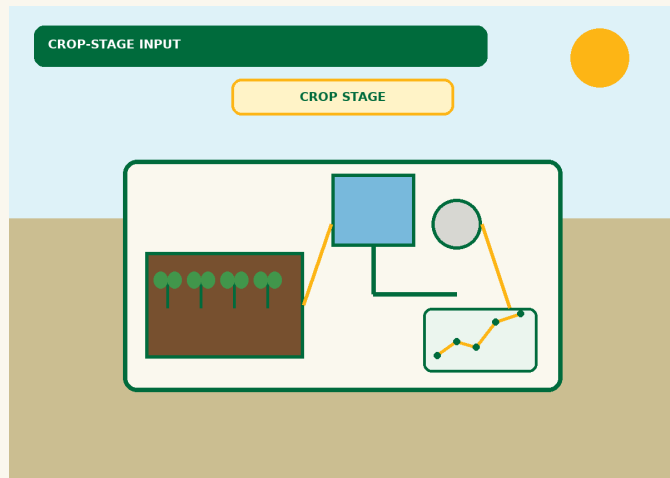
TECHNICAL CONCEPT

Weather observations can help estimate atmospheric water demand and recent water supply for irrigation decisions.

ANALYSIS / DESIGN TASK

Interpret a simplified weather dataset.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



CROP STAGE

CROP-STAGE INPUT

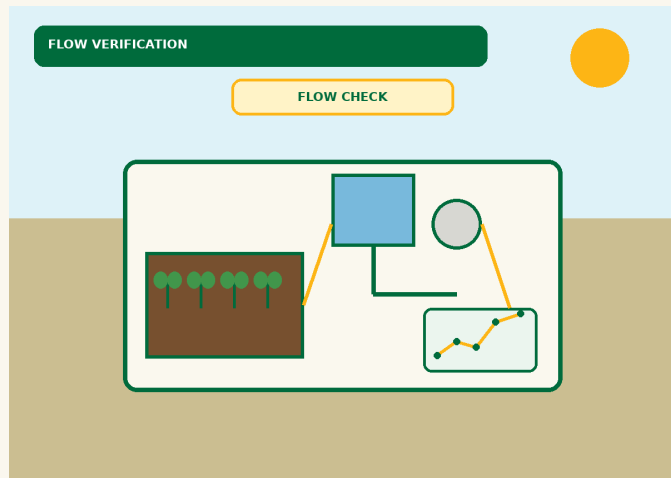
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Crop development stage influences rooting, canopy, sensitivity, and water demand and can modify control rules.

ANALYSIS / DESIGN TASK

Adjust a rule for two crop stages.



FLOW CHECK

FLOW VERIFICATION

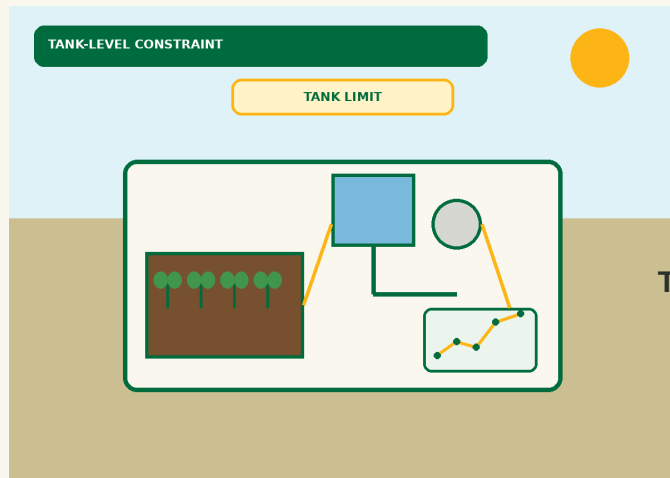
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Flow measurement verifies whether commanded irrigation actually delivered water at the expected rate.

ANALYSIS / DESIGN TASK

Compare commanded and measured delivery.



TANK LIMIT

TANK-LEVEL CONSTRAINT

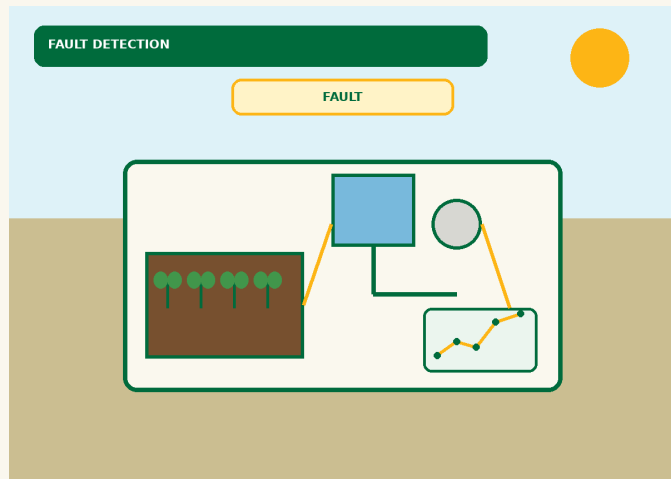
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Tank level can constrain irrigation decisions when stored water is insufficient for the requested operation.

ANALYSIS / DESIGN TASK

Apply a minimum-reserve rule.



FAULT

FAULT DETECTION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

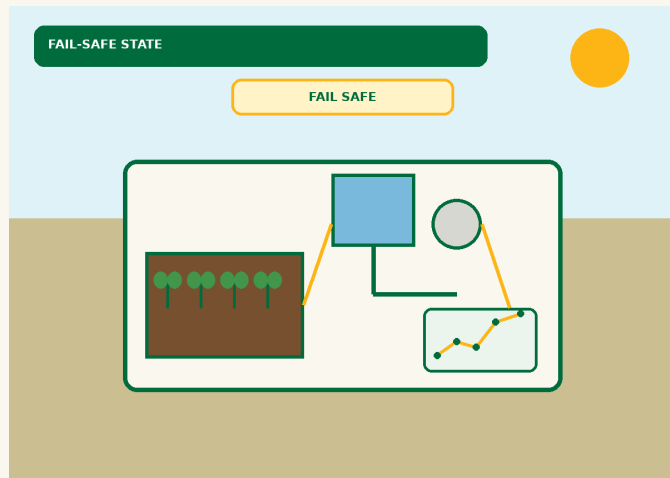
TECHNICAL CONCEPT

Fault detection identifies abnormal behavior such as no flow after valve opening, leakage, sensor failure, or unexpected tank decline.

ANALYSIS / DESIGN TASK

Diagnose one fault signature.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



FAIL SAFE

FAIL-SAFE STATE

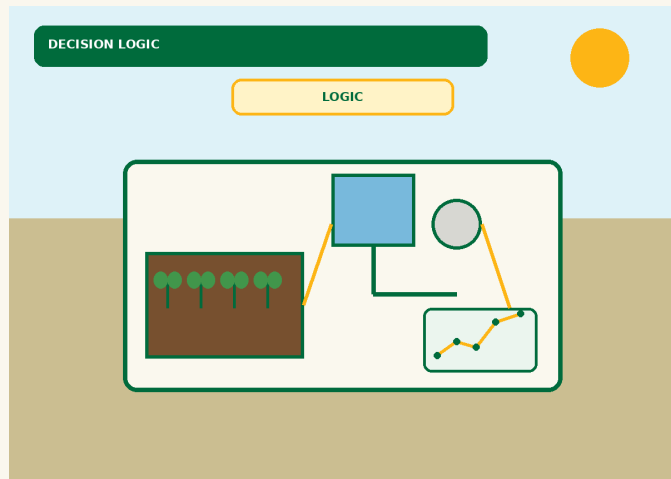
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A fail-safe state is the predefined safe operating condition used when critical measurements, power, or control functions fail.

ANALYSIS / DESIGN TASK

Specify a safe response to sensor failure.



LOGIC

DECISION LOGIC

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

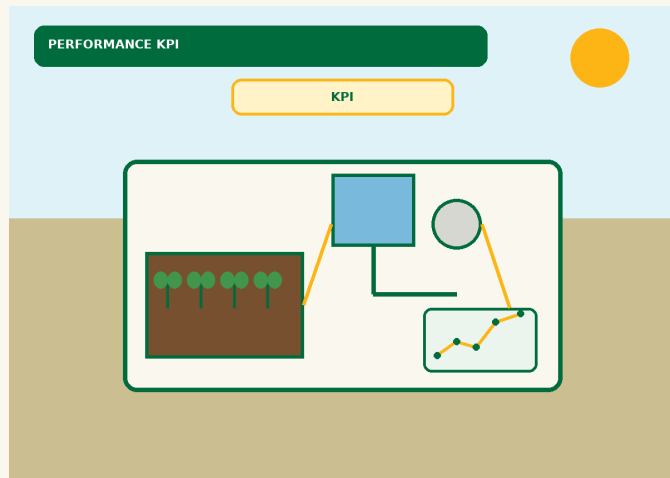
TECHNICAL CONCEPT

Decision logic combines states, thresholds, priorities, constraints, and actions into an explicit operational rule set.

ANALYSIS / DESIGN TASK

Write pseudocode for a simple controller.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



KPI

PERFORMANCE KPI

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A key performance indicator quantifies control performance, such as water applied, threshold violations, uniformity, or crop response.

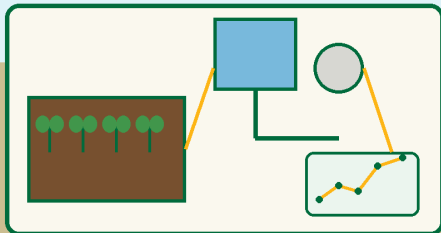
ANALYSIS / DESIGN TASK

Select three KPIs.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

RULE-BASED BENCHMARK

BENCHMARK



BENCHMARK

RULE-BASED BENCHMARK

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

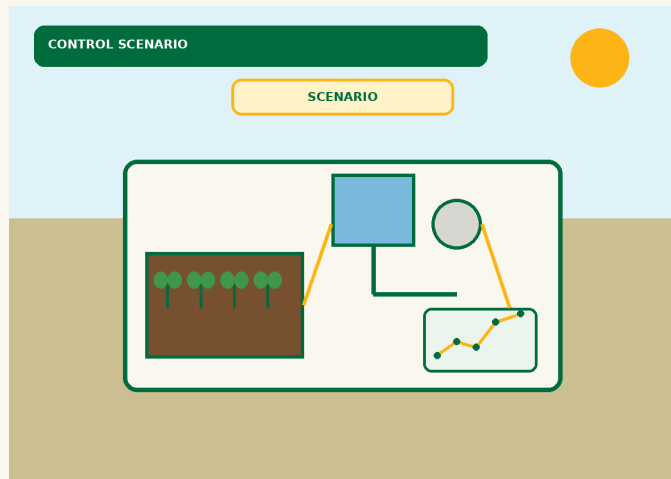
TECHNICAL CONCEPT

A benchmark controller provides a reference strategy against which a more advanced control approach can be evaluated.

ANALYSIS / DESIGN TASK

Define a fixed-schedule benchmark.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



SCENARIO

CONTROL SCENARIO

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Control scenarios test system behavior under different water supply, demand, sensor, or operational conditions.

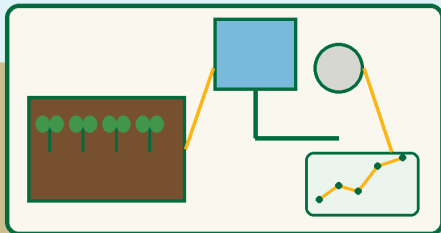
ANALYSIS / DESIGN TASK

Compare normal and constrained scenarios.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

SMART IRRIGATION EVALUATION

EVALUATE



EVALUATE

SMART IRRIGATION EVALUATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Evaluation compares water use, crop-zone conditions, reliability, faults, and operational performance against a benchmark.

ANALYSIS / DESIGN TASK

Determine whether adaptive control improved the defined KPIs.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE